AS · Unit 2 · WCH12 Energetics, Group Chemistry, Halogenoalkanes and Alcohols

Intermolecular forces

Covalent bonds hold the atoms together inside a molecule. Much weaker forces act between neighbouring molecules, and those forces decide whether a substance is a gas, a liquid or a solid, how hot it must be to boil, and what it will dissolve in. This topic explains the three kinds of intermolecular force, how to tell which ones a substance has, and how to use them to explain real data.

SPEC 7.1 The three types of intermolecular force

WHAT THIS SPEC POINT ASKS

Understand the nature of (i) London forces (instantaneous dipole–induced dipole), (ii) permanent dipole–permanent dipole interactions and (iii) hydrogen bonds.

Intermolecular is not intramolecular

- **Intramolecular** bonds are the covalent bonds **inside** a molecule. They are strong.
- **Intermolecular forces** act **between** separate molecules. They are much weaker (a hydrogen bond is roughly a tenth of a covalent bond, London forces weaker still).
- When a simple molecular substance melts or boils, **only the intermolecular forces are overcome**. The covalent bonds stay intact: boiling water gives H_2O molecules, not H and O atoms.

(i) London forces

1. Electrons in a molecule are always moving. At any instant they may be **unevenly spread**, giving an **instantaneous (temporary) dipole**.
2. That dipole repels or attracts the electrons of a neighbouring molecule, **inducing a dipole** in it.
3. The δ+ end of one attracts the δ− end of the other. The dipoles keep vanishing and re-forming, but on average there is an attraction.

- London forces act between **all** molecules (and atoms of noble gases), polar or not.
- They get **stronger with more electrons per molecule**: more electrons give bigger instantaneous dipoles.
- They are also stronger when molecules have **more points of contact** (long, straight molecules can lie close along their length).
- Other names: dispersion forces, van der Waals forces, instantaneous dipole–induced dipole forces.

(ii) Permanent dipole–permanent dipole interactions

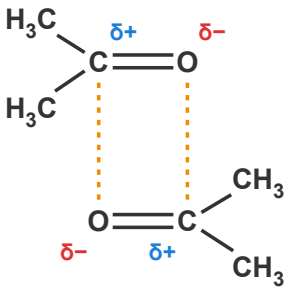
- A molecule is **polar** if it contains polar bonds (atoms of different electronegativity) **and** its shape doesn't cancel them out. HCl, CH_3Cl and propanone are polar; CO_2 and CCl_4 are not (symmetric, so the dipoles cancel).
- The δ+ end of one polar molecule attracts the δ− end of the next. This acts **in addition to** London forces.
- For molecules with a similar number of electrons, pd–pd forces make the polar one boil higher: propanone (32 electrons, polar) boils at 56 °C; butane (34 electrons, non-polar) at about 0 °C.

hydrogen chloride

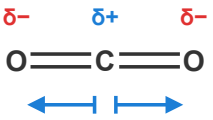


δ− of one molecule attracts δ+ of the next

propanone



carbon dioxide



bond dipoles cancel:
a non-polar molecule

Orange dotted lines: permanent dipole–dipole attractions between polar molecules. CO_2 has polar C=O bonds, but its linear shape makes the two dipoles cancel, so it has London forces only.

(iii) Hydrogen bonds

- A hydrogen bond forms between a **δ+ hydrogen** that is covalently bonded to **N, O or F**, and a **lone pair on an N, O or F atom** of another molecule.
- N, O and F are very electronegative and small, so the N–H, O–H and F–H bonds are very polar, and the hydrogen (with no inner electrons) is a tiny, very exposed δ+ centre.
- The three atoms X–H···Y lie in a **straight line (180°)**, because the lone pair and the bonding pair repel each other as far apart as possible.
- Hydrogen bonds are the **strongest** intermolecular force, but still much weaker than covalent bonds.

KEY IDEA

Strength order (for molecules of similar size): hydrogen bonds > permanent dipole–dipole > London forces. But a big molecule's London forces can outweigh a small molecule's hydrogen bonds: I_2 (106 electrons, London only) is a solid, while water is a liquid.

MARK-SCHEME WORDING

- London forces: "**instantaneous / temporary dipole**" caused by "an **uneven / asymmetrical distribution of electrons**", which "**induces a dipole**" in a neighbouring molecule (WCH12 Oct 2024). ACCEPT van der Waals, dispersion.

- Strength: "(hydrogen iodide has) **more electrons**". IGNORE "bigger molecule"; DO NOT AWARD with wrong extra reasons such as "more polarising".
- Comparisons must be comparative: "hydrogen bonds are **stronger than** London forces". "Hydrogen bonding is strong" alone scores nothing (WCH12 Jun 2022).
- Levels answers: "F_2 has **only** London forces; HCl has pd–pd as well; CH_3OH has hydrogen bonds" (WCH12 Jun 2021). Say "only" where it applies.
- Any mention of **breaking covalent bonds** on boiling loses the mark.

Often asked as: explain how London forces arise; a 6-mark comparison of boiling or liquefaction temperatures (F_2, HCl, CH_3OH; H_2 and NH_3); which forces are present in a named substance.

COMMON MISTAKE

Describing the trend in London forces down a group when asked **how they arise** (WCH12 Oct 2024 examiner report).

Comparing M_r instead of **number of electrons** (WCH12 Jan 2025 examiner report).

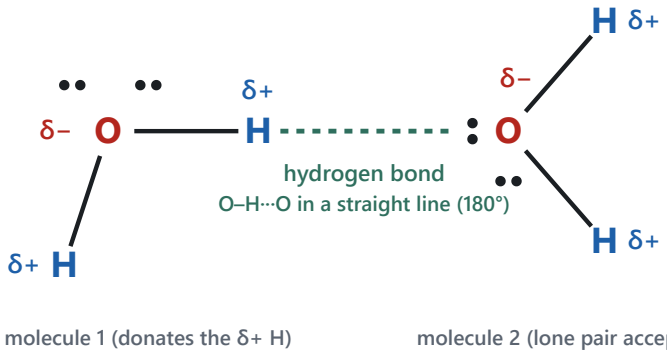
Calling the covalent N–H bond a "hydrogen bond", or putting a hydrogen bond **inside** one molecule (WCH12 Jan 2022 examiner report).

Forgetting that polar and hydrogen-bonded molecules **also** have London forces.

SPEC 7.2 Hydrogen bonding in water, ammonia and hydrogen fluoride

WHAT THIS SPEC POINT ASKS

Understand the interactions in H_2O, liquid NH_3 and liquid HF that lead to hydrogen bonding.

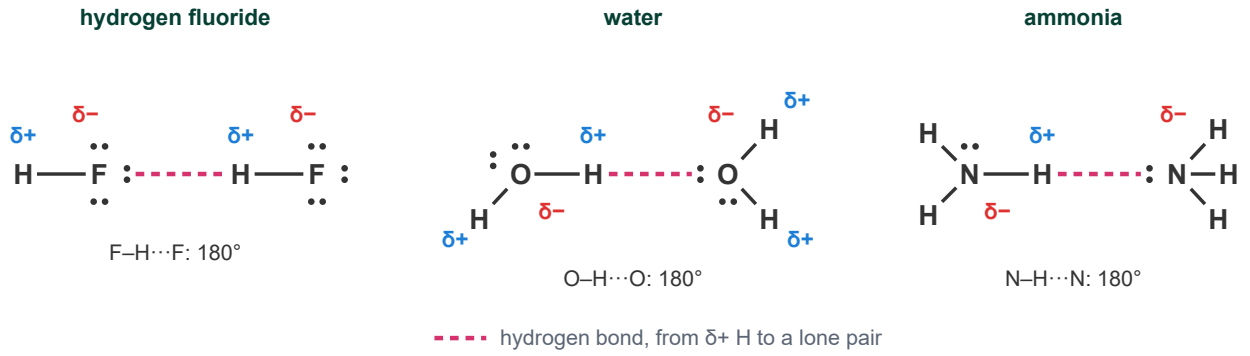


A hydrogen bond (dotted) runs from the δ+ H of one water molecule to a lone pair on the O of the next. O–H···O is a straight line.

What each molecule brings

Molecule	δ+ H atoms (donors)	Lone pairs on N/O/F (acceptors)	Average hydrogen bonds per molecule
H_2O	2	2 (on O)	**2**
NH_3	3	1 (on N)	1 (limited by the single lone pair)
HF	1	3 (on F)	1 (limited by the single H)

- Each hydrogen bond needs one δ+ H **and** one lone pair. Water is the only one of the three with equal numbers (two of each), so it forms the **most hydrogen bonds per molecule**. That's why water boils highest of the three.
- **HF** forms the **strongest single** hydrogen bond, because F is the most electronegative element, so H–F is the most polar bond. In liquid HF the molecules form zig-zag chains, F–H···F–H···F.
- **NH_3**: N is less electronegative than O or F, so its hydrogen bonds are weaker. In liquid ammonia each N lone pair accepts an H from a neighbouring molecule.



In each case the hydrogen bond (pink) runs from a δ+ hydrogen to a lone pair on N, O or F, in a straight line. Only one hydrogen bond is drawn in each pair; real liquids form a network of them.

Drawing a hydrogen bond (the marks)

1. Draw both molecules with all atoms (no "O_2H").
2. Put **δ+ on the H** and **δ− on the O (N, F)**.
3. Show the **lone pair** on the accepting O, N or F.
4. Draw the hydrogen bond as a **dotted or dashed line** from that lone pair to the H.
5. Make O–H···O **straight** and label it **180°**.

MARK-SCHEME WORDING

- "Hydrogen bond from a **lone pair on the oxygen** of one molecule to the **(δ+) hydrogen** of another", "**linear** O–H···O, labelled **180°**", "δ+ on H, δ− on O". The bond must be dotted, not solid (WCH12 Oct 2019).
- Conditions: "H attached to N, O or F" plus "N, O or F with a lone pair".
- HF strongest: "F has the highest electronegativity, so H–F is the most polar bond" (WCH12 Jan 2020).
- Ammonia: "hydrogen bonds from the **N lone pair** to the δ+ H of a **different** NH_3 molecule" (WCH12 Jun 2023).

Often asked as: draw a hydrogen bond between two water molecules, two alcohol molecules, or ethanol and water; explain why HF boils higher than HCl.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

Water forms more hydrogen bonds than NH_3 or HF because it has **two lone pairs and two H atoms** (a 2 : 1 ratio compared with NH_3's one lone pair) (WCH12 Jan 2025).

The N–H···N angle in ammonia need not be drawn as exactly 180° (WCH12 Jun 2023).

COMMON MISTAKE

Missing lone pairs, missing the 180° label, or drawing H–O···H (wrong atoms lined up) (WCH12 Jan 2025 examiner report).

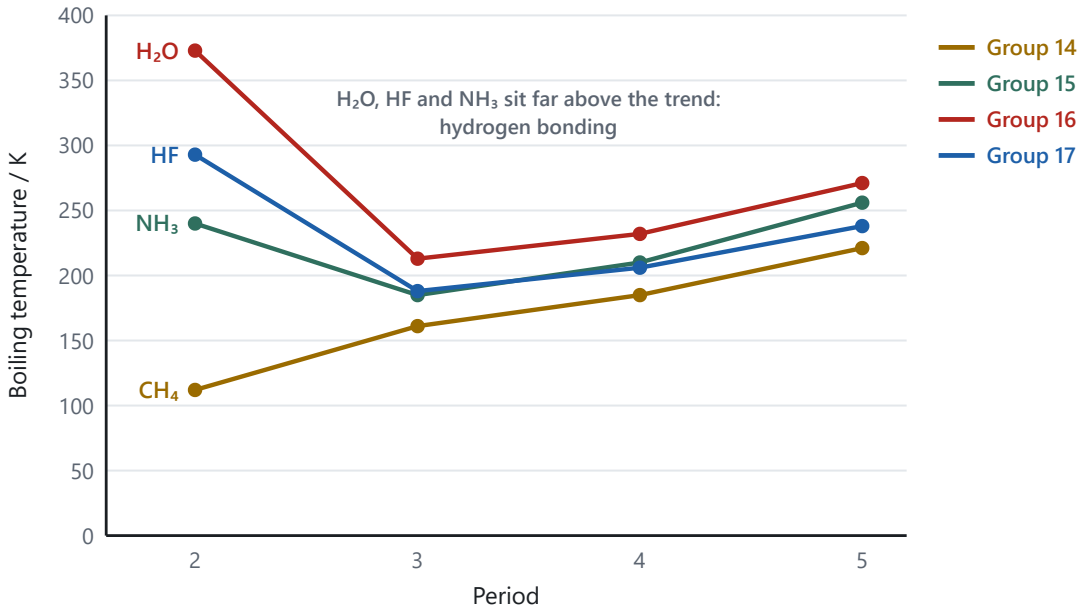
Drawing the hydrogen bond to the δ+ H of a C–H group: C–H is not polar enough.

SPEC 7.3 The anomalous properties of water

WHAT THIS SPEC POINT ASKS

Understand how hydrogen bonding explains (i) water's high melting and boiling temperatures compared with similar molecules and (ii) why ice is less dense than liquid water.

(i) High melting and boiling temperatures



From period 3 down, boiling temperature rises as the number of electrons (and so London forces) rises. H_2O, HF and NH_3 break the pattern because of hydrogen bonding. CH_4 (Group 14) has no hydrogen bonding and follows the trend.

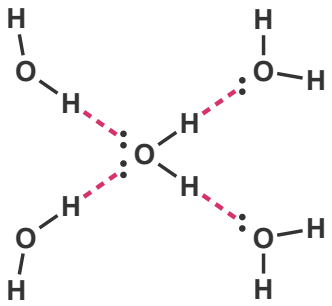
- Going down Group 16 from H_2S to H_2Te, boiling temperature rises because the molecules have **more electrons**, so **stronger London forces**.
- Extending that trend back, H_2O "should" boil at around −80 °C. It actually boils at **100 °C**.

- Water molecules form **hydrogen bonds** (two per molecule), which are **much stronger** than the London and pd–pd forces in H_2S. **Much more energy** is needed to separate the molecules.
- The same reasoning explains why water is a liquid, not a gas, at room temperature, and why its melting temperature (0 °C) is so high for its size.

(ii) Ice is less dense than water

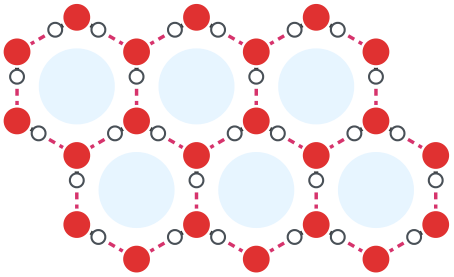
- In **ice**, every water molecule is hydrogen-bonded to **four** others in a fixed, tetrahedral arrangement. This makes an **open lattice** with a lot of empty space between the molecules.
- When ice melts, some hydrogen bonds **break**. The open structure collapses and the molecules can move **closer together**, so liquid water is **denser** (about 1.00 g cm^{-3}, compared with about 0.92 g cm^{-3} for ice).
- So ice **floats**. Ponds freeze from the top down, and the ice layer insulates the water below, so aquatic life survives.
- For almost every other substance, the solid is denser than the liquid.

one molecule, four hydrogen bonds



in three dimensions the four bonds point to the corners of a tetrahedron

oxygen atoms in ice: an open lattice



red: O · white: H · pink dashes: hydrogen bonds
the pale circles mark the empty space in each ring

Left: in ice each molecule donates two hydrogen bonds and accepts two. Right: the hydrogen bonds lock the oxygen atoms into hexagonal rings with a lot of empty space (each oxygen also bonds to a molecule above or below the page). Melting breaks some hydrogen bonds and the molecules move closer together.

Why ice floats

1. In ice, every water molecule is hydrogen-bonded to four others, arranged tetrahedrally around it: two bonds through its own hydrogens and two through the lone pairs on its oxygen.
2. The hydrogen bonds have a fixed length and direction, so they hold the molecules in a rigid, **open lattice** with a lot of empty space, and the molecules further apart than in liquid water.

3. When ice melts, some hydrogen bonds break, the open lattice collapses, and the molecules can pack closer together. So the same mass takes up less volume: liquid water is denser than ice.

   Ice floating has a big consequence: lakes freeze from the top down, and the ice layer insulates the water below, so aquatic life survives the winter.

MARK-SCHEME WORDING

Water v ammonia (WCH12 Jan 2025, three marks):

- "similar number of electrons, so London forces are similar"
- "water forms **more hydrogen bonds** (per molecule) than ammonia"
- "density of water increases as it changes from solid to liquid" because the **open hydrogen-bonded structure of ice** collapses.

ACCEPT "hydrogen bonds in water are stronger because O is more electronegative than N". IGNORE comments on pd–pd forces.

Often asked as: explain water's boiling temperature compared with H_2S or NH_3; explain the density change on melting.

COMMON MISTAKE

Quoting M_r; saying ammonia can't hydrogen bond; comparing the density of water with the density of ammonia instead of ice with liquid water (WCH12 Jan 2025 examiner report: half the candidates scored zero).

Saying covalent O–H bonds break when water boils.

SPEC 7.4 Predicting hydrogen bonding in other molecules

WHAT THIS SPEC POINT ASKS

Predict whether hydrogen bonding is present in molecules similar to H_2O, NH_3 and HF.

The two-part test

A substance hydrogen-bonds **with itself** only if its molecules have **both**:

1. an **H atom bonded directly to N, O or F** (an O–H, N–H or H–F group), **and**
2. a **lone pair on an N, O or F** atom.

Molecule	O–H / N–H present?	Hydrogen bonds between its own molecules?
Alcohols, e.g. CH_3OH, C_2H_5OH	O–H	**yes**
Carboxylic acids, e.g. CH_3COOH	O–H	**yes** (strongly: they can pair up as dimers)
Amines, e.g. CH_3NH_2	N–H	**yes**
Urea, $(NH_2)_2CO$	N–H	**yes** (and pd–pd)
Propane-1,2,3-triol (glycerol)	three O–H	**yes**, three per molecule: very viscous
Propanone, CH_3COCH_3	no (only C–H)	**no**: pd–pd and London only
Ethers, aldehydes, esters	no	**no** with themselves
Halogenoalkanes, e.g. CH_3Cl	no	**no**: pd–pd and London
HCl, HBr, HI	H–X, but X is not N, O or F	**no**

- Molecules like **propanone** can't hydrogen-bond to each other, but the lone pairs on their O **can accept** a hydrogen bond from water. That's why propanone mixes with water (7.6).
- **Count the hydrogen-bonding groups.** More O–H groups → more hydrogen bonds per molecule → higher boiling temperature and viscosity (ethane-1,2-diol and glycerol boil much higher than ethanol).

MARK-SCHEME WORDING

- Urea: "urea forms **hydrogen bonds**", "(and permanent) dipole–dipole forces", "these are **stronger than London forces**" (WCH12 Jan 2022).
- Propan-2-ol v propanone: "propan-2-ol forms hydrogen bonds (between its molecules); propanone has only London and pd–pd forces; hydrogen bonds are stronger". Both dissolve in water because both can form hydrogen bonds **with water** (WCH12 Oct 2025).
- Glycerol: "forms **three** hydrogen bonds per molecule" / "three OH groups" (WCH12 Jun 2023).

Often asked as: which of these molecules can form hydrogen bonds; compare the boiling temperatures of an alcohol and a ketone or alkane; explain the properties of a molecule in a Section C context.

COMMON MISTAKE

Drawing a hydrogen bond from an H bonded to carbon.

Forgetting London forces in a comparison: very few answers mention them (WCH12 Jan 2022 examiner report).

Saying HCl hydrogen-bonds. Cl is not N, O or F.

SPEC 7.5 Intermolecular forces and physical properties

WHAT THIS SPEC POINT ASKS

Explain, using intermolecular forces: (i) boiling temperatures of alkanes as chain length increases, (ii) the effect of branching, (iii) the low volatility of alcohols compared with alkanes with a similar number of electrons, and (iv) the boiling temperatures of HF to HI.

The answer pattern

Every boiling-point answer has the same three links:

1. **Name the force(s)** present in each substance.
2. **Compare** their strength, with a reason (number of electrons, contact area, hydrogen bonding).
3. **More energy is needed to overcome** the stronger intermolecular forces (to **separate the molecules**), so the boiling temperature is higher.

(i) Chain length in alkanes

Alkane	CH_4	C_2H_6	C_3H_8	C_4H_{10}	C_5H_{12}	C_6H_{14}
Boiling temperature / °C	−162	−89	−42	−1	36	69

- Each extra CH_2 adds **8 electrons** and makes the molecule longer, so there are **more electrons** and **more points of contact**. London forces get stronger, so more energy is needed to separate the molecules.
- Alkanes are non-polar: London forces are the **only** intermolecular force.

(ii) Branching

Isomer of C_5H_{12}	pentane	2-methylbutane	2,2-dimethylpropane
Shape	straight chain	one branch	most branched, nearly spherical
Boiling temperature / °C	36	28	10

- Isomers have the **same number of electrons**, so the difference is all about shape.
- Straight chains can lie **close together along their length**: **more points of contact / larger surface area of contact**, so stronger London forces.
- Branched molecules are more compact and can't get as close, so they have fewer points of contact, weaker London forces and a lower boiling temperature.

(iii) Alcohols v alkanes

Compound	Electrons	Forces	Boiling temperature / °C
propane, C_3H_8	26	London only	−42
ethanol, C_2H_5OH	26	London + **hydrogen bonds** (+ pd–pd)	78
butane, C_4H_{10}	34	London only	−1
propan-1-ol, C_3H_7OH	34	London + **hydrogen bonds** (+ pd–pd)	97

- With the **same number of electrons**, the London forces are **similar**. The alcohol also has **hydrogen bonds** between its O–H groups, which are **much stronger**, so it needs much more energy to boil. Alcohols are **less volatile** (evaporate less readily).

(iv) The hydrogen halides

	HF	HCl	HBr	HI
Electrons	10	18	36	54
Boiling temperature / °C	**20**	−85	−67	−35

- **HCl → HBr → HI**: boiling temperature **rises** because the number of electrons rises, so **London forces get stronger**. (The permanent dipole actually gets **smaller** from HCl to HI, as electronegativity falls, so the trend is driven by London forces, not pd–pd.)
- **HF** is far higher than expected because only HF forms **hydrogen bonds**, which are much stronger than the forces between HCl molecules.

Explaining the boiling temperatures of HF to HI

1. **HCl → HBr → HI:** the number of electrons increases (18, 36, 54), so the London forces between the molecules get stronger.

2. The molecules actually become **less** polar down the group, because the difference in electronegativity between H and the halogen gets smaller. So the permanent dipole–dipole forces get weaker.

3. The boiling temperature still rises, so the increase in London forces must outweigh the decrease in dipole–dipole forces. More energy is needed to separate the molecules.

4. **HF** has the fewest electrons, yet it has the highest boiling temperature, because HF molecules form **hydrogen bonds** (H bonded to F, with lone pairs on F). Hydrogen bonds are much stronger than the other intermolecular forces, so far more energy is needed to overcome them.

MARK-SCHEME WORDING

- Alkanes: "London forces **increase with the number of electrons**", "more energy needed to **overcome / break these intermolecular forces**". ALLOW "more points of contact / surface area". Mentioning hydrogen bonds or pd–pd loses the force marks; mentioning breaking covalent bonds loses the energy mark (WCH12 Oct 2020).
- Branching: "London forces are **stronger between straight chains**" because of "**greater surface area / more points of contact**". DO NOT AWARD longer carbon bonds or breaking covalent bonds (WCH12 Oct 2019).
- Hydrogen halides: "London forces increase from HCl to HI (more electrons)", "**only HF** has hydrogen bonding", "hydrogen bonds are (much) stronger" (WCH12 Jun 2022). DO NOT AWARD ions, or breaking H–Cl bonds.
- Group 15 hydrides: London forces rise with electrons; "(only) ammonia has hydrogen bonding" (WCH12 Jun 2025). DO NOT AWARD "London forces are similar down the group".

Often asked as: 6-mark comparisons of boiling temperatures; explain a data table or a graph of boiling temperature against period or chain length.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

Iodine v bromine: "iodine has **more electrons per molecule**", so stronger London forces, so a higher melting temperature. DO NOT AWARD "between atoms" (WCH12 Jun 2021).

1-iodopropane has the strongest London forces of the halogenopropanes even though its C–I dipole is the smallest (WCH12 Oct 2021).

Pentane is a **liquid**, so part of the energy from burning it isn't measured in $\Delta_c H$ calculated from gaseous bond enthalpies; boiling and combustion trends must not be confused (WCH12 Oct 2020).

COMMON MISTAKE

"Straight-chain isomers boil higher" with no reason; "London forces are present in both" with no comparison (WCH12 Oct 2019 examiner report).

Talking about size, not the **number of electrons** (WCH12 Oct 2021 examiner report).

Claiming hydrogen bonds in HCl, HBr, HI or in 1-iodopropane.

SPEC 7.6 Choosing a solvent

WHAT THIS SPEC POINT ASKS

Understand why: (i) water dissolves some ionic compounds (hydration of ions); (ii) water dissolves simple alcohols (hydrogen bonding); (iii) water is a poor solvent for compounds that can't hydrogen-bond with it, including polar ones like halogenoalkanes; (iv) non-aqueous solvents dissolve compounds with similar intermolecular forces.

The energy rule behind dissolving

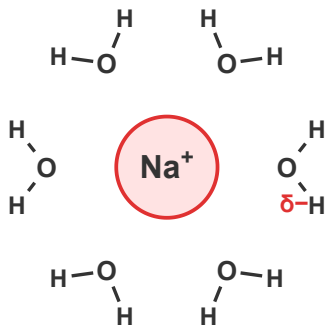
To dissolve, you must **break** the forces between solute particles **and** some of the forces between solvent molecules. That costs energy. It is paid back by the **new attractions between solute and solvent**. A substance dissolves well when the new solute–solvent attractions are **similar in strength to (or stronger than)** the ones broken.

"Like dissolves like" is the short version, but on its own it earns nothing: always **name the forces**.

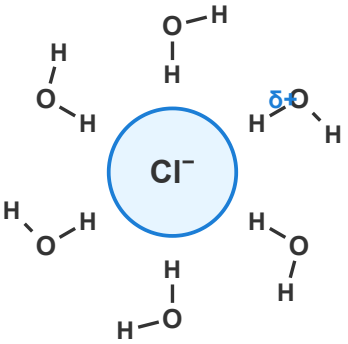
(i) Ionic compounds in water: hydration

- Water is polar. Its **δ− oxygen** is attracted to **positive ions**; its **δ+ hydrogens** are attracted to **negative ions**.
- Several water molecules surround each ion. These are **hydrated ions**, and forming them releases energy (**hydration enthalpy**).
- If the energy released by hydration is enough to compensate for breaking up the **ionic lattice**, the compound dissolves.
- If the lattice is very strong (small or highly charged ions, e.g. $MgCO_3$, $CaCO_3$, $BaSO_4$), hydration can't make up for it and the compound is **insoluble**.

When you draw hydrated ions: at least two water molecules round each ion, **O pointing at the cation**, **H pointing at the anion**. These are **ion–dipole** attractions, not hydrogen bonds.



δ− oxygen atoms face the positive ion



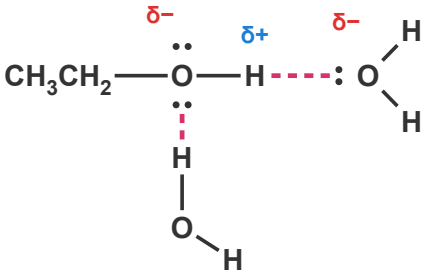
δ+ hydrogen atoms face the negative ion

Hydrated ions. The water molecules turn so that their oppositely charged end faces the ion. The attraction between an ion and these polar molecules releases the energy that breaks up the lattice.

(ii) Simple alcohols in water

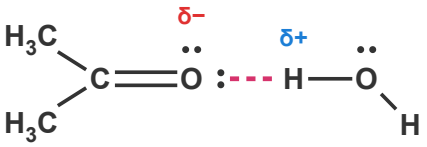
- Methanol, ethanol and propan-1-ol mix with water in all proportions. The **O–H group forms hydrogen bonds with water molecules**, and these new hydrogen bonds are similar in strength to the ones broken in pure water and pure alcohol.
- As the hydrocarbon chain gets longer, solubility **falls**. The non-polar chain can only form London forces with water, and it gets in the way of water's own hydrogen bonding. Hexan-1-ol is only slightly soluble.

ethanol and water



ethanol both gives and accepts hydrogen bonds

propanone and water



propanone can only accept a hydrogen bond

Hydrogen bonds (pink) between water and organic molecules. Ethanol can form them both ways, through its O–H hydrogen and through its oxygen lone pairs. Propanone has no O–H, but water can still hydrogen bond to its oxygen, which is why propanone mixes with water.

(iii) Water as a poor solvent

- **Halogenoalkanes** are polar (C–X dipole) but **can't form hydrogen bonds** with water. To dissolve, they would have to break water's strong hydrogen bonds and replace them with weaker pd–pd and London forces. So

they are **immiscible** with water and form a separate layer.

- The same applies to **alkanes** and other non-polar compounds, which can only offer London forces.
- That's why halogenoalkane hydrolysis with silver nitrate is done in **ethanol**: ethanol mixes with both the halogenoalkane and the aqueous silver nitrate (10.8).

(iv) Non-aqueous solvents

- A non-polar solvent such as **hexane** or **cyclohexane** dissolves non-polar solutes (iodine, alkanes, fats, halogenoalkanes): solute and solvent both rely on **London forces**, so the new attractions match the old ones.
- **Iodine**: barely soluble in water, very soluble in cyclohexane, where it gives a **purple** solution.
- Ionic compounds don't dissolve in hexane: London forces with hexane release nowhere near enough energy to break the lattice.

Solute	Water	Ethanol	Hexane / cyclohexane
NaCl (ionic)	dissolves (hydration)	very slightly	no
Ethanol	mixes (hydrogen bonds)	—	dissolves (London forces with its chain)
1-bromobutane	immiscible	mixes	mixes
Iodine	very slightly (pale yellow/brown)	dissolves (brown)	dissolves (purple)

MARK-SCHEME WORDING

- Hydration diagram: "the **oxygen** is closest to the (calcium) **cation**; the **hydrogen** is closest to the (chloride) **anion**", at least **two** water molecules per ion (WCH12 Jun 2021). DO NOT AWARD O_2H. Penalise a labelled "hydrogen bond" or full charges on water (WCH12 Oct 2024).
- Insoluble ionic solid: "**large** amount of energy needed to **break the ionic lattice**" but "**smaller** amount released on **hydration**" (WCH12 Jun 2019). DO NOT AWARD molecules, atoms or London forces.
- Iodine in cyclohexane: "cyclohexane and iodine form **London forces**"; "hydrogen bonds between water molecules are stronger than the London forces between iodine and water" (WCH12 Jun 2019).
- Ethanol and water mix because "they form hydrogen bonds with each other", "from the lone pair on an O of one to the δ+ H of the other", and the new forces are **similar in strength** to those broken (WCH12 Jun 2021).

Often asked as: draw ions hydrated by water; explain why a solid is insoluble; why iodine dissolves in cyclohexane but not water; why ethanol and water mix.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

Layer colours when iodine is shaken with water and cyclohexane: **pale yellow** aqueous layer, **pink / purple / violet** organic layer (WCH12 Jun 2019).

Explaining insolubility by reversing the dissolving argument: compare solute–solute, solvent–solvent and solute–solvent forces (WCH12 Jun 2019 examiner report).

COMMON MISTAKE

"Like dissolves like" with no forces named (WCH12 Jun 2019, Oct 2019 examiner reports).

Saying ions form hydrogen bonds with water, or writing about intermolecular forces in an ionic solid.

Saying iodine forms hydrogen bonds with cyclohexane, or "reacts" with it.

Calling the aqueous layer colourless when "most", not all, of the iodine moves into the organic layer.

Quick checks

QUICK CHECK

What happens when liquid bromine, Br_2, boils?

- Br–Br covalent bonds break, forming bromine atoms.
- London forces between Br_2 molecules are overcome.
- Hydrogen bonds between Br_2 molecules are broken.
- Permanent dipole–dipole forces between Br_2 molecules are overcome.

Boiling separates whole molecules, so no covalent bonds break. Br_2 is non-polar, so the only intermolecular forces are London forces.

QUICK CHECK

Why do the boiling temperatures of the noble gases increase from helium to xenon?

- The atoms become more polar.
- The covalent bonds between the atoms get stronger.
- The atoms have more electrons, so the instantaneous and induced dipoles are larger and the London forces stronger.
- Heavier atoms move more slowly, so they cannot escape from the liquid.

Noble gas atoms are single, non-polar atoms with no covalent bonds between them. Their only attraction is London forces, which increase with the number of electrons.

QUICK CHECK

Which of these substances forms hydrogen bonds between its own molecules?

- fluoromethane, CH_3F
- propanone, CH_3COCH_3
- methylamine, CH_3NH_2
- hydrogen sulfide, H_2S

Methylamine has H atoms bonded to nitrogen, which also carries a lone pair. In CH_3F and propanone no H is bonded to N, O or F; sulfur is not electronegative enough for H_2S to hydrogen bond.

QUICK CHECK

Which statement about hydrogen bonding in ice is correct?

- Each water molecule is hydrogen-bonded to two others.
- The hydrogen bonds hold the molecules further apart than in liquid water, so ice is less dense.
- The O–H···O angle at each hydrogen bond is 104.5°.
- When ice melts, the covalent O–H bonds break.

In ice each molecule is hydrogen-bonded to four others in an open tetrahedral lattice. The O–H···O arrangement is linear (180°); 104.5° is the H–O–H bond angle inside one molecule. Melting only breaks some hydrogen bonds.

QUICK CHECK

HCl boils at 188 K and HBr at 206 K. Which explanation is correct?

- HBr is more polar, so its dipole–dipole forces are stronger.
- HBr forms hydrogen bonds but HCl does not.
- The H–Br bond is stronger than the H–Cl bond.
- HBr has more electrons, so its London forces are stronger, outweighing its smaller dipole.

Bromine is less electronegative than chlorine, so HBr is actually less polar. Neither forms hydrogen bonds, and bond strength inside the molecule is irrelevant to boiling. The extra electrons strengthen the London forces enough to raise the boiling temperature.

QUICK CHECK

Butane (34 electrons) boils at −1 °C and propan-1-ol (34 electrons) at 97 °C. What is the main reason?

- Propan-1-ol molecules form hydrogen bonds; butane molecules only have London forces.
- Propan-1-ol has more electrons, so stronger London forces.
- Butane is branched, so its molecules have less contact.
- Propan-1-ol has stronger covalent bonds.

The two have the same number of electrons, so their London forces are similar. The difference is the hydrogen bonding through the –OH group, which needs much more energy to overcome. Butane is a straight chain, not branched.

QUICK CHECK

Why is 1-chlorobutane almost insoluble in water, even though its molecules are polar?

- It reacts with water.
- Its molecules are too large to fit between water molecules.
- It cannot form hydrogen bonds with water, and its weaker attractions to water do not make up for breaking hydrogen bonds between water molecules.
- Its C–Cl bond is non-polar.

To dissolve, it would have to break into water's hydrogen-bonded network. The dipole–dipole and London attractions it can form with water are too weak to compensate, so it stays as a separate layer.

QUICK CHECK

In a solution of sodium chloride, how are the water molecules arranged around the ions?

- H atoms point towards Na^+; O atoms point towards Cl^-.
- O atoms point towards Na^+; H atoms point towards Cl^-.
- O atoms point towards both ions.
- Water molecules form covalent bonds to both ions.

The oxygen of water is δ−, so it is attracted to the positive sodium ion; the hydrogens are δ+, so they are attracted to the negative chloride ion. These ion–dipole attractions are what hydration means.